Ecommerce Customer Churn Analysis

1. Project Description:

a) Problem Statement –

The E-commerce company is experiencing the user churn which is threatening our recurring revenue and long-term market share. Therefore, the problem here is to understand why customers are leaving an online e-commerce company.

b) Project Overview –

This project analyses customer attrition for an e-commerce platform across 5,630 customer records to identify why users discontinue service. Using MySQL for data cleaning, transformation, and aggregations alongside Power BI for interactive visualization, it identifies key attrition drivers such as tenure drop-offs, unresolved support complaints, and delivery distances. The resulting dashboards deliver actionable, data-driven retention strategies to reduce the platform's 16.8% churn rate and increase customer lifetime value.

2. Dataset Overview:

Columns names –

Identifier & Target – CustomerID, Churn.

Customer Demographics – Gender, MaritalStatus, CityTier, NumberOfAddress.

Customer Lifecycle & Engagement – Tenure, HourSpendOnApp, NumberOfDeviceRegistered, PreferredLoginDevice.

Purchasing Behaviour – OrderCount, DaySinceLastOrder, PreferedOrderCat, PreferredPaymentMode

Monetization & Incentives – CashbackAmount, CouponUsed, OrderAmountHikeFromlastYear.

Logistics & Customer Experience – WarehouseToHome, Complain, SatisfactionScore.

No. of Rows – 5,630

No. of Columns – 20

Data Source – Kaggle datasets.

3. Tech Stack Used:

MySQL - Using MySQL for data cleaning, transformation, and data analysis.

Power BI - Using Power BI for Data Visualization.

4. Data Cleaning:

Before embarking on any analysis, it is essential to ensure the dataset is clean and reliable. The data cleaning process involves handling missing values, correcting inconsistencies, and formatting the data for analysis. In this project, we carefully cleaned the dataset to ensure the accuracy and integrity of our findings.

1. Checking for duplicate rows

```
with duplicates as (select customerid ,row_number() over(partition by customerid ) as rn from ecommercechurn)
select * from duplicates where rn > 1;
```

Result Grid | Filter Rows:

customerid	rn

The query returns an empty table, showing there are no duplicate rows.

2. Checking for null values

CouponUsed, DaysSinceLastOrder, HourSpendOnApp, OrderAmountHikeFromLastYear, OrderCount, Tenure, and WarehouseToHome all have null values present, and the number of null values present for each column can be seen in the below table.

```sql
SELECT 'Tenure' as ColumnName, COUNT(*) AS NullCount
FROM ecommercechurn
WHERE Tenure IS NULL
UNION
SELECT 'WarehouseToHome' as ColumnName, COUNT(*) AS NullCount
FROM ecommercechurn
WHERE warehousetohome IS NULL
UNION
SELECT 'HourSpendonApp' as ColumnName, COUNT(*) AS NullCount
FROM ecommercechurn
WHERE hourspendonapp IS NULL
UNION
SELECT 'OrderAmountHikeFromLastYear' as ColumnName, COUNT(*) AS NullCount
FROM ecommercechurn
WHERE orderamounthikefromlastyear IS NULL
UNION
SELECT 'CouponUsed' as ColumnName, COUNT(*) AS NullCount
FROM ecommercechurn
WHERE couponused IS NULL
UNION
SELECT 'OrderCount' as ColumnName, COUNT(*) AS NullCount
FROM ecommercechurn
WHERE ordercount IS NULL
UNION
SELECT 'DaySinceLastOrder' as ColumnName, COUNT(*) AS NullCount
FROM ecommercechurn
WHERE daysincelastorder IS NULL ;
```

	ColumnName	NullCount
1	CouponUsed	256
2	DaySinceLastOrder	307
3	HourSpendonApp	255
4	OrderAmountHikeFromLastYear	265
5	OrderCount	258
6	Tenure	264
7	WarehouseToHome	251

3. Handling null values

```sql
SELECT AVG(Hourspendonapp) FROM ecommercechurn;
UPDATE ecommercechurn
SET Hourspendonapp = 2.9315
WHERE Hourspendonapp IS NULL ;

SELECT AVG(tenure) FROM ecommercechurn;
UPDATE ecommercechurn
SET tenure = 10.1899
WHERE tenure IS NULL ;

SELECT AVG(orderamounthikefromlastyear) FROM ecommercechurn;
UPDATE ecommercechurn
SET orderamounthikefromlastyear = 15.7079
WHERE orderamounthikefromlastyear IS NULL ;

SELECT  AVG(WarehouseToHome) FROM ecommercechurn;
UPDATE ecommercechurn
SET WarehouseToHome = 15.6399
WHERE WarehouseToHome IS NULL ;

SELECT AVG(couponused) FROM ecommercechurn;
UPDATE ecommercechurn
SET couponused = 1.7510
WHERE couponused IS NULL ;

SELECT AVG(ordercount) FROM ecommercechurn;
UPDATE ecommercechurn
SET ordercount = 3.0080
WHERE ordercount IS NULL ;

SELECT AVG(daysincelastorder) FROM ecommercechurn;
UPDATE ecommercechurn
SET daysincelastorder = 4.5435
WHERE daysincelastorder IS NULL ;
```

ColumnName	NullCount
Tenure	0
WarehouseToHome	0
HourSpendonApp	0
OrderAmountHikeFromLastYear	0
CouponUsed	0
OrderCount	0
DaySinceLastOrder	0

In order to handle these null values, we will fill the nulls with the mean value of their respective columns. Now that the null values have been replaced with their mean, there are no more null values left.

4. Creating a new column from an already existing “churn” column

While going through the dataset, we noticed that the churn column contained 0 and 1 values. 0 means that a customer did not churn, while 1 means that a customer churned. It is difficult to remember which is for what, so we will create a new column called ‘CustomerStatus’ that shows “Stayed” when a customer did not churn and “Churned” when a customer churned.

```
alter table ecommercechurn
add column customerstatus varchar(50) after `churn`;

update ecommercechurn
set customerstatus = case when churn = 0 then 'Stayed' when churn = 1 then 'Churned' end ;
select distinct customerstatus from ecommercechurn;
alter table ecommercechurn drop column churn;
```

customerstatus
Churned
Stayed

The new column “CustomerStatus” has two distinct values called "churned" and "stayed," as earlier described.

5. Creating a new column from an already existing “complain” column

Very similar to what we did in the section above, we noticed that the complain column also contained 0 and 1. ‘0’ means that the customer did not record any complaints, while ‘1’ means the customer recorded a complaint. For clarity purposes, we will create a new column called ‘ComplainRecieved’ that shows “No” when a customer did not complain and “Yes” when a customer complained.

```
ALTER TABLE ecommercechurn
ADD column ComplainRecieved VARCHAR(10) after `customerstatus`;

UPDATE ecommercechurn
SET ComplainRecieved =
CASE  WHEN complain = 1 THEN 'Yes' WHEN complain = 0 THEN 'No' END;
select distinct complainrecieved from ecommercechurn;
alter table ecommercechurn drop column complain;
```

complainrecieved
Yes
No

The new column “complainrecieved” has two distinct values called “yes” and “no,” as earlier described.

6. Checking values in each column for correctness and accuracy

```sql
select distinct customerstatus from ecommercechurn;
select distinct ComplainRecieved from ecommercechurn;
select distinct Tenure from ecommercechurn;
select distinct PreferredLoginDevice from ecommercechurn;
select distinct CityTier from ecommercechurn;
select distinct WarehouseToHome from ecommercechurn;
select distinct PreferredPaymentMode from ecommercechurn;
select distinct Gender from ecommercechurn;
select distinct HourSpendOnApp from ecommercechurn;
select distinct NumberOfDeviceRegistered from ecommercechurn;
select distinct PreferedOrderCat from ecommercechurn;
select distinct SatisfactionScore from ecommercechurn;
select distinct MaritalStatus from ecommercechurn;
select distinct NumberOfAddress from ecommercechurn;
select distinct OrderAmountHikeFromlastYear from ecommercechurn;
select distinct CouponUsed from ecommercechurn;
select distinct OrderCount from ecommercechurn;
select distinct DaySinceLastOrder from ecommercechurn;
select distinct CashbackAmount from ecommercechurn;
```

After going through each column, we noticed some redundant values in some columns and a wrongly entered value. This will be explored and fixed.

6.1 Fixing redundancy in “PreferedLoginDevice” Column

Notice that phone and mobile phone appear in the column, but they mean the same thing. So we will replace the mobile phone with phone.

```sql
-- Replace mobile phone with phone
UPDATE ecommercechurn
SET preferredlogindevice = 'phone'
WHERE preferredlogindevice = 'mobile phone';
```

6.2 Fixing redundancy in “PreferedOrderCat” Column

Notice that mobile phone and phone appear in the column, but they mean the same thing. So we will replace the phone with mobile phone as the category name.

```sql
-- Replace mobile with mobile phone
UPDATE ecommercechurn
SET preferedordercat = 'Mobile Phone'
WHERE Preferedordercat = 'Mobile';
```

6.3 Fixing redundancy in “PreferredPaymentMode” Column

Notice that Cash on Delivery and COD both appear in the column, but they mean the same thing. So we will replace COD with Cash on Delivery.

```
--  Replace cod with Cash on Delivery
UPDATE ecommercechurn
SET PreferredPaymentMode  = 'Cash on Delivery'
WHERE PreferredPaymentMode  = 'COD';
```

6.4 Fixing wrongly entered values in “WarehouseToHome” column

Notice the 126 and 127 values; they are definitely outliers and most likely wrongly entered. To fix this, we will correct the values to 26 and 27 to fall within the range of the values found in this column.

```
-- Replace value 126 with 26
UPDATE ecommercechurn
SET warehousetohome = '26'
WHERE warehousetohome = '126';

-- Replace value 127 with 27
UPDATE ecommercechurn
SET warehousetohome = '27'
WHERE warehousetohome = '127';
```

5. Data Exploration And Data Analysis:

Answering business questions.

1. What is the overall customer churn rate?

```
select customerstatus,count(customerid),
100 * count(customerid)/(select count(CustomerID) from ecommercechurn)  as rate
from ecommercechurn
where customerstatus = 'churned'
group by customerstatus;
```

customerstatus	count(customerid)	rate
Churned	948	16.8384

The churn rate of 16.84% indicates that a significant portion of customers in the dataset have ended their association with the company.

2. How does the churn rate vary based on the preferred login device?

```
select PreferredLoginDevice,
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
count(customerid),
round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID),2)  as rate
from ecommercechurn
group by 1;
```

PreferredLoginDevice	totalchurnedcustomers	count(customerid)	rate
phone	624	3996	15.62
Computer	324	1634	19.83

The preferred login device appears to have some influence on customer churn rates. Customers who prefer logging in using a computer have a slightly higher churn rate compared to customers who prefer logging in using their phones. This may indicate that customers who access the platform via a computer might have different usage patterns, preferences, or experiences that contribute to a higher likelihood of churn. Understanding these preferences can help businesses optimize their platform and user experience for different login devices, ensuring a seamless and engaging experience for customers.

3. What is the distribution of customers across different city tiers?

```
SELECT citytier,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1;
```

citytier	count(customerid)	totalchurnedcustomers	rate
3	1722	368	21.37
1	3666	532	14.51
2	242	48	19.83

City Tier 1 is typically a major metropolitan area with the highest level of economic development and infrastructure. These cities are usually the most populous and have significant commercial and business centres.

City Tier 2 is considered smaller or secondary urban centres compared to Tier 1 cities.

City Tier 3 is further down the hierarchy and generally refers to smaller towns or cities with a smaller population and less developed infrastructure compared to Tier 1 and 2 cities.

The result suggests that the city tier has an impact on customer churn rates. Tier 1 cities have a relatively lower churn rate compared to Tier 2 and Tier 3 cities. This could be attributed to various factors such as competition, customer preferences, or the availability of alternatives in different city tiers.

4. Is there any correlation between the warehouse-to-home distance and customer churn?

```
alter table ecommercechurn
add column warehousetohomerange varchar(50) after warehousetohome;

update  ecommercechurn
set warehousetohomerange = case when warehousetohome <= 10 then 'very close distance'
when warehousetohome >10 and warehousetohome <=20 then 'close distance'
when warehousetohome >20 and warehousetohome <=30 then 'moderate distance'
when warehousetohome > 30 then 'far distance'
end;

select warehousetohomerange,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

warehousetohomerange	count(customerid)	totalchurnedcustomers	rate
far distance	469	98	20.90
moderate distance	874	176	20.14
close distance	2318	408	17.60
very close distance	1969	266	13.51

The distance between the warehouse and the customer's home seems to have some influence on customer churn rates. Customers residing in closer proximity to the warehouse tend to have lower churn rates, while customers living at further distances are more likely to churn. This suggests that factors such as delivery times, shipping costs, or convenience may play a role in customer retention. The company can utilize these insights to optimize its logistics and delivery strategies, ensuring better service for customers residing at further distances and implementing retention initiatives for customers in higher churn rate categories.

5. Which is the most preferred payment mode among churned customers?

```
SELECT PreferredPaymentMode,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

PreferredPaymentMode	count(customerid)	totalchurnedcustomers	rate
Cash on Delivery	514	128	24.90
E wallet	614	140	22.80
CC	273	59	21.61
UPI	414	72	17.39
Debit Card	2314	356	15.38
Credit Card	1501	193	12.86

The most preferred payment mode among churned customers is cash on delivery. The preferred payment mode seems to have some influence on customer churn rates. Payment modes such as “Cash on Delivery” and “E-wallet” show higher churn rates, indicating that customers using these payment modes are more likely to churn. On the other hand, payment modes like “Credit Card” and “Debit Card” have relatively lower churn rates.

6. What is the typical tenure for churned customers?

```
ALTER TABLE ecommercechurn
ADD TenureRange VARCHAR(50) after tenure;

UPDATE ecommercechurn
SET TenureRange =
CASE
    WHEN tenure <= 6 THEN '6 Months'
    WHEN tenure > 6 AND tenure <= 12 THEN '1 Year'
    WHEN tenure > 12 AND tenure <= 24 THEN '2 Years'
    WHEN tenure > 24 THEN 'more than 2 years'
END;

select tenurerange,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

tenurerange	count(customerid)	totalchurnedcustomers	rate
6 Months	2150	697	32.42
1 Year	1584	156	9.85
2 Years	1467	95	6.48
more than 2 years	429	0	0.00

This shows that customers who have been with the company for longer periods, specifically more than 2 years in this case, have shown a lower likelihood of churn compared to customers in shorter tenure groups.

7. Is there any difference in churn rate between male and female customers?

```
select gender,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

gender	count(customerid)	totalchurnedcustomers	rate
Male	3384	600	17.73
Female	2246	348	15.49

Both male and female customers exhibit churn rates, with males having a slightly higher churn rate compared to females. However, the difference in churn rates between the genders is relatively small. This suggests that gender alone may not be a significant factor in predicting customer churn.

8. How does the average time spent on the app differ for churned and non-churned customers?

```
SELECT customerstatus, avg(hourspendonapp) AS AverageHourSpentonApp
FROM ecommercechurn
GROUP BY customerstatus;
```

customerstatus	AverageHourSpentonApp
Churned	2.9641
Stayed	2.9287

Both churned and staying customers have the same average hours spent on the app; this indicates that the average app usage time does not seem to be a differentiating factor between customers who churned and those who stayed.

9. Does the number of registered devices impact the likelihood of churn?

```
select numberofdeviceregistered,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

numberofdeviceregistered	count(customerid)	totalchurnedcustomers	rate
6	162	56	34.57
5	881	198	22.47
4	2377	392	16.49
3	1699	254	14.95
2	276	26	9.42
1	235	22	9.36

There seems to be a correlation between the number of devices registered by customers and the likelihood of churn. Customers with a higher number of registered devices, such as 6 or 5, exhibit higher churn rates. On the other hand, customers with fewer registered devices, such as 2 or 1, show relatively lower churn rates.

10. Which order category is most preferred among churned customers?

```
select PreferedOrderCat,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

PreferedOrderCat	count(customerid)	totalchurnedcustomers	rate
Mobile Phone	2080	570	27.40
Fashion	826	128	15.50
Laptop & Accessory	2050	210	10.24
Others	264	20	7.58
Grocery	410	20	4.88

The analysis suggests that different order categories have varying impacts on customer churn rates. Customers who primarily order items in the “Mobile Phone” category have the highest churn rate, indicating a potential need for

targeted retention strategies for this group. On the other hand, the “Grocery” category exhibits the lowest churn rate, suggesting that customers in this category may have higher retention and loyalty.

11. Is there any relationship between customer satisfaction scores and churn?

```
select SatisfactionScore,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

SatisfactionScore	count(customerid)	totalchurnedcustomers	rate
5	1108	264	23.83
3	1698	292	17.20
4	1074	184	17.13
2	586	74	12.63
1	1164	134	11.51

The result indicates that customers with higher satisfaction scores, particularly those who rated their satisfaction as 5, have a relatively higher churn rate compared to other satisfaction score categories. This suggests that even highly satisfied customers may still churn, highlighting the importance of proactive customer retention strategies across all satisfaction levels.

12. Does the marital status of customers influence churn behaviour?

```
select MaritalStatus,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

MaritalStatus	count(customerid)	totalchurnedcustomers	rate
Single	1796	480	26.73
Divorced	848	124	14.62
Married	2986	344	11.52

Single customers have the highest churn rate compared to customers with other marital statuses. This indicates that single customers may be more likely to discontinue their relationship with the company. On the other hand, married customers have the lowest churn rate, followed by divorced customers.

13. How many addresses do churned customers have on average?

```
SELECT round(AVG(numberofaddress),0) AS Averagenumofchurnedcustomeraddress
FROM ecommercechurn
WHERE customerstatus = 'churned';
```

Averagenumofchurnedcustomeraddres
4

On average, customers who churned had four addresses associated with their accounts.

14. Do customer complaints influence churned behavior?

```
select ComplainRecieved,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

ComplainRecieved	count(customerid)	totalchurnedcustomers	rate
Yes	1604	508	31.67
No	4026	440	10.93

The fact that a larger proportion of customers who stopped using the company's services, registered complaints. It indicates the importance of dealing with and resolving customer concerns. This is vital for decreasing the number of customers who leave and building lasting loyalty. By actively listening to customer feedback, addressing their issues, and consistently working on improving the quality of their offerings, companies can create a better overall experience for customers. This approach helps to minimize the chances of customers leaving and fosters stronger relationships with them in the long run.

15. How does the use of coupons differ between churned and non-churned customers?

```
with cte as (
SELECT customerstatus, SUM(couponused) AS SumofCouponUsed
FROM ecommercechurn
GROUP BY customerstatus)
select customerstatus,sumofcouponused,sum(sumofcouponused) over() as total,
100 * sumofcouponused/sum(sumofcouponused) over() as rate
from cte ;
```

customerstatus	sumofcouponused	total	rate
Churned	1630	9922	16.4281
Stayed	8292	9922	83.5719

The higher coupon usage among stayed customers indicates their higher level of loyalty and engagement with the company. By implementing strategies to reward loyalty, provide personalized offers, and maintain continuous engagement, the company can further leverage coupon usage as a tool to strengthen customer loyalty and increase overall customer retention.

16. What is the average number of days since the last order for churned customers?

```
SELECT Round(AVG(daysincelastorder),0) AS AverageNumofDaysSinceLastOrder
FROM ecommercechurn
WHERE customerstatus = 'churned';
```

AverageNumofDaysSinceLastOrder
3

The fact that churned customers have, on average, only had a short period of time since their last order indicates that they recently stopped engaging with the company. By focusing on enhancing the overall customer experience, implementing targeted retention initiatives, and maintaining continuous engagement, the company can work towards reducing churn and increasing customer loyalty.

17. Is there any correlation between cashback amount and churn rate?

```sql
ALTER TABLE ecommercechurn
ADD column cashbackamountrange VARCHAR(50);

UPDATE ecommercechurn
SET cashbackamountrange =
CASE
    WHEN cashbackamount <= 100 THEN 'Low Cashback Amount'
    WHEN cashbackamount > 100 AND cashbackamount <= 200 THEN 'Moderate Cashback Amount'
    WHEN cashbackamount > 200 AND cashbackamount <= 300 THEN 'High Cashback Amount'
    WHEN cashbackamount > 300 THEN 'Very High Cashback Amount'
END;

select cashbackamountrange,count(customerid),
sum( case when customerstatus = 'churned' then 1 else 0 end) as totalchurnedcustomers,
Round(100 * sum( case when customerstatus = 'churned' then 1 else 0 end)/count(CustomerID) ,2) as rate
from ecommercechurn
group by 1
order by rate desc;
```

cashbackamountrange	count(customerid)	totalchurnedcustomers	rate
Moderate Cashback Amount	4305	814	18.91
High Cashback Amount	1157	124	10.72
Very High Cashback Amount	156	10	6.41
Low Cashback Amount	12	0	0.00

Customers who received moderate cashback amounts had a relatively higher churn rate, while those who received higher and very high cashback amounts exhibited lower churn rates. Customers who received lower cashback amounts also had a 100% retention rate. This suggests that offering higher cashback amounts can positively influence customer loyalty and reduce churn.

Insight Section

- The dataset includes 5,630 customers, providing a substantial sample size for analysis.
- The overall churn rate is 16.84%, indicating significant customer attrition.
- Customers who prefer logging in with a computer have slightly higher churn rates compared to phone users, suggesting different usage patterns and preferences.

- Tier 1 cities have lower churn rates than Tier 2 and Tier 3 cities, possibly due to competition and customer preferences.
- Proximity to the warehouse affects churn rates, with closer customers showing lower churn, highlighting the importance of optimizing logistics and delivery strategies.
- “Cash on Delivery” and “E-wallet” payment modes have higher churn rates, while “Credit Card” and “Debit Card” have lower churn rates, indicating the influence of payment preferences on churn.
- Longer tenure is associated with lower churn rates, emphasizing the need for building customer loyalty early on.
- Male customers have slightly higher churn rates than female customers, although the difference is minimal.
- App usage time does not significantly differentiate between churned and non-churned customers.
- More registered devices correlate with higher churn rates, suggesting the need for consistent experiences across multiple devices.
- “Mobile Phone” order category has the highest churn rate, while “Grocery” has the lowest, indicating the importance of tailored retention strategies for specific categories.
- Highly satisfied customers (rating 5) have a relatively higher churn rate, highlighting the need for proactive retention strategies at all satisfaction levels.
- Single customers have the highest churn rate, while married customers have the lowest, indicating the influence of marital status on churn.
- Churned customers have an average of four associated addresses, suggesting higher mobility.

- Customer complaints are prevalent among churned customers, emphasizing the importance of addressing concerns to minimize churn.

- Coupon usage is higher among non-churned customers, showcasing the effectiveness of loyalty rewards and personalized offers.

- Churned customers have had a short time since their last order, indicating recent disengagement and the need for improved customer experience and retention initiatives.

- Moderate cashback amounts correspond to higher churn rates, while higher amounts lead to lower churn, suggesting the positive impact of higher cashback on loyalty.

6. Recommendations:

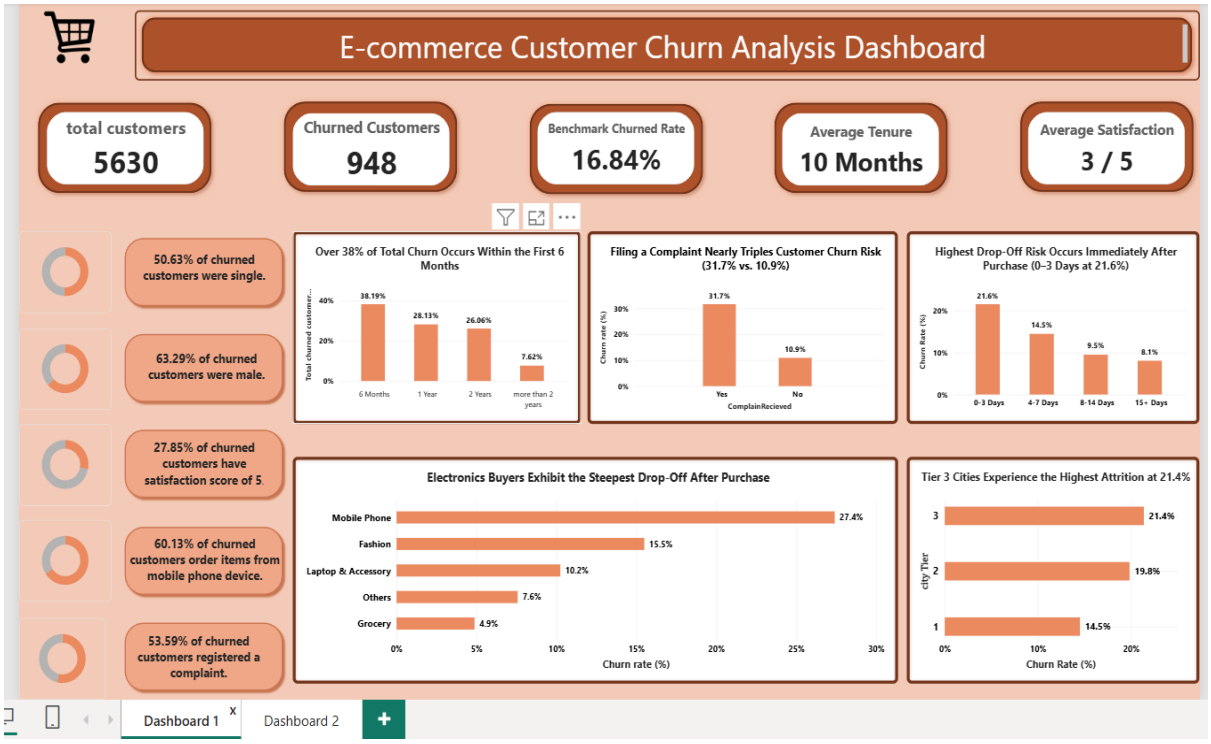
1. Enhance the user experience for customers who prefer logging in via computer. Conduct research to identify and address any issues they might be facing, making improvements to ensure a smoother and more enjoyable experience.
2. Tailor retention strategies based on the different city tiers. Understand the preferences and needs of customers in each tier to provide targeted offerings and incentives that resonate with them.
3. Optimize logistics and delivery to improve customer satisfaction. Focus on reducing delivery times, lowering shipping costs, and finding ways to make the process more convenient, especially for customers living further away.
4. Simplify payment processes, particularly for options like “Cash on Delivery” and “E-wallet.” Enhance security measures and offer incentives for customers to use more reliable payment methods such as “Credit Card” and “Debit Card.”
5. Improve customer support and complaint resolution. Address customer complaints promptly and effectively, providing satisfactory resolutions. Actively listen to customer feedback, make necessary improvements, and demonstrate a commitment to addressing their concerns.
6. Develop targeted retention initiatives for specific order categories, such as the “Mobile Phone” category. Offer exclusive discounts, rewards, or promotions to incentivize continued engagement and loyalty
7. Ensure a consistent experience across different devices. Implement features like cross-device syncing, personalized recommendations, and easy account management to encourage usage and retention across multiple devices.

8. Proactively engage and reward satisfied customers across all satisfaction levels. Regularly communicate with them through personalized messages, exclusive offers, and loyalty programs to maintain their loyalty and reduce the risk of churn.
9. Consider increasing cashback incentives to retain customers, especially those who are more likely to churn. Conduct A/B testing to determine the most effective cashback levels that encourage higher customer retention rates.

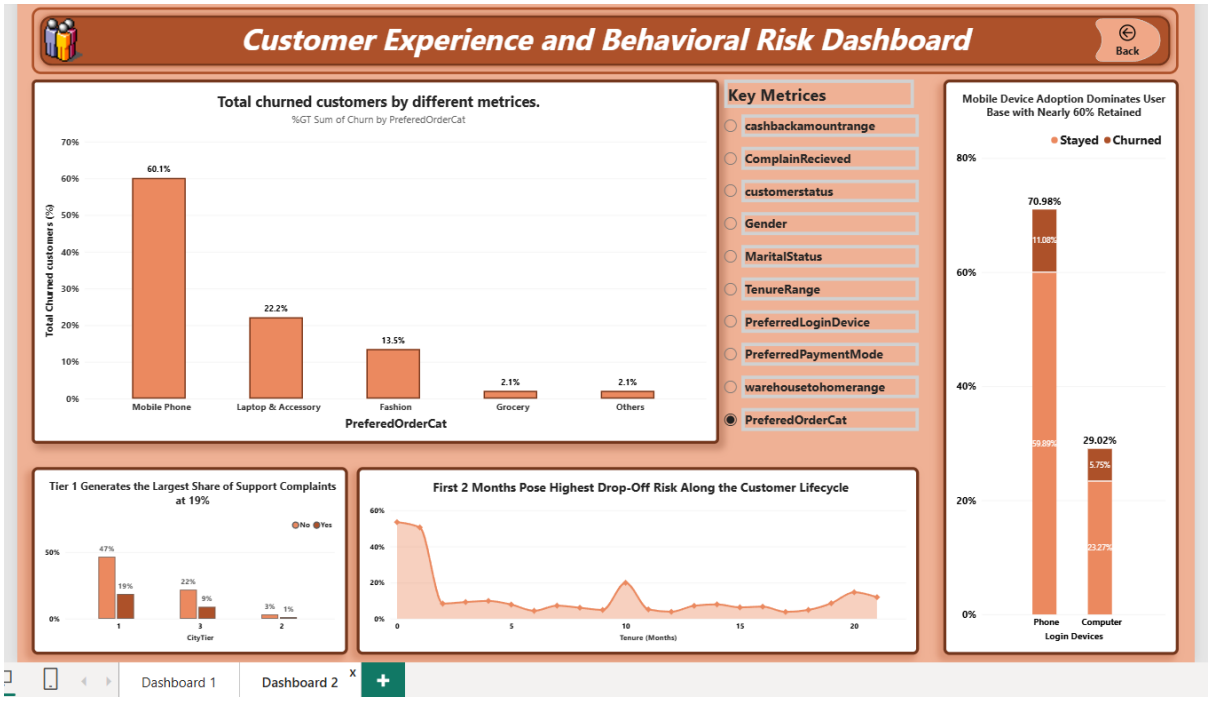
By implementing these recommendations, this company can improve customer retention, reduce churn rates, and build long-term loyalty, leading to sustainable growth and success.

7. Data Visualization:

Power BI dashboard 1 –



Power BI dashboard 2 –



End